

The E-Management Systems Impact on the Digital Public Services: The Case Study of X-Road of Estonia

تأثير أنظمة الإدارة الإلكترونية في الخدمات العامة الرقمية: دراسة حالة X-Road في إستونيا

Abderrahmane Yessad¹, Feriel Kerrat²

¹University of Mostaganem (Algeria), yessad.abderrahmane@univ-mosta.dz
Laboratoire Dynamique Macro-Économique Et Changements Structurels

²University of Mostaganem (Algeria), feriel.kerrat@univ-mosta.dz
Laboratoire Dynamique Macro-Économique Et Changements Structurels

Received: 14/03/2026

Accepted: 17/06/2026

Published:01/07/2026

Abstract:

The research investigates the role of electronic management systems in optimizing the governance of digital public services. Adopting a descriptive-analytical methodology, the study focuses on Estonia's X-Road system, a global benchmark in digital governance. The findings demonstrate that Estonia's integrated e-governance framework acts as a significant economic engine, contributing over 2% to the national GDP annually through enhanced operational speed and administrative efficiency. The practical implications of this digital maturity are profound: high-stakes transactions, such as vehicle registration, are executed remotely in under 15 minutes, while tax filings are typically completed within five minutes. Most notably, the system has streamlined the democratic process, reducing the average time for online voting to a mere 90 seconds. Ultimately, this study underscores how digital transformation serves as a cornerstone for modern and high-performance public administration.

Keywords:E-management, digital public services, digital transformation.

JELClassificationCodes : H11, L86, O33

ملخص:

تهدف هذه الدراسة إلى تسليط الضوء على دور نظم الإدارة الإلكترونية في إدارة الخدمات الرقمية العمومية، مستعملين في الدراسة المنهج الوصفي التحليلي ودراسة حالة لنظام X-Road في إستونيا كدولة رائدة في هذا المجال. من خلال هذه الدراسة، تبين أن أنظمة الإدارة الإلكترونية في حكومة إستونيا

¹ Corresponding author: Feriel Kerrat, e-mail:feriel.kerrat@univ-mosta.dz

تساهم سنوياً أكثر من 2 في المائة من ناتجها المحلي الإجمالي بفضل السرعة في التطبيق والكفاءة في الأداء. على سبيل المثال يمكن شراء سيارة في إستونيا عن بعد في أقل من 15 دقيقة، ويستغرق تقديم إقرار ضريبي عبر الإنترنت لشخصاً عادياً لا يزيد عن خمس دقائق، وتستغرق المشاركة في الانتخابات عن طريق التصويت عبر الإنترنت 90 ثانية في المتوسط.

كلمات مفتاحية: الإدارة الالكترونية، الخدمات الالكترونية العمومية، التحول الرقمي، نظام **X-Road**.

تصنيفات JEL : H11، L86، O33.

1. INTRODUCTION

Digital transformation serves as a fundamental force for human and socio economic development. This paradigm shift gained significant momentum post the COVID-19 pandemic, which acted as a global inflection point, mandating the rapid acceleration, exploration, and implementation of digital frameworks across the globe. Where the mature nations were significantly better positioned to provide targeted assistance to the citizens compared to their counterparts that lacked such foundational digital resources.(UNDP, 2024). At its core, electronic management systems (EMS) function as an integrative architecture, synthesizing traditionally siloed operations, ranging from document and project management to human resources, supply chain logistics, and customer relationship management (CRM) – into a unified, digital ecosystem.

By facilitating seamless connectivity between government agencies and private enterprises, this interoperability platform serves as a foundational layer for the rapid deployment of innovative services, products, and emerging business models. Beyond simple technical integration, such a framework cultivates a dynamic economy.(Mustafa, 2020).

Prominent ecosystems such as the X-Road illustrate this transformative approach. These systems demonstrate how standardized digital foundations can empower nations to achieve digital sovereignty while fostering a collaborative global environment for innovation.(David & Sechi , 2022). Furthermore, Estonia has achieved global recognition as the world's most advanced digital society. This distinction underscores the nation's success in fostering a comprehensive digital ecosystem.(Techforgood, 2022). Consequently, Estonia stands as a preeminent global exemplar of a fully realized e-society. Ultimately, the

Estonian model demonstrates that digital maturity is as much a social and institutional accomplishment as it is a technological one.(Sandra & Raul, 2016).

Research question: Previous information leads to a fundamental question: What mechanisms have enabled the Estonian government to leverage e-management systems to optimize administrative functions and fortify its economic sustainability in the digital transformation era?

Research Objectives :

1. To evaluate the conceptual and structural evolution of e-management systems within the paradigm of the digital transformation era.
2. To assess the strategic opportunities and systemic challenges inherent in the implementation of e-management.
3. To examine how Estonia's e-management framework serves as a global benchmark for sustainable e-governance and national competitiveness.

1-The digital transformation:

1-1-Overview: Digital transformation encompasses a holistic integration of advanced technologies across all institutional strata, fundamentally redefining operational architectures and the mechanisms through which organizations deliver value to their stakeholders.

While the strategic implementation of digital transformation manifests uniquely across diverse industrial sectors, the underlying structural evolution can be distilled into four distinct developmental phases, which function as a staged progression, enabling a systematic transition from legacy operations to fully integrated digital ecosystems:(COURSERA, 2023).

- **Early Stage:** The initial trajectory is defined by meticulous strategic planning, the cultivation of essential cross-sectoral partnerships, and a rigorous assessment of the organization's baseline digital capabilities.
- **Mid-Stage:** During this transition, leadership prioritizes high-impact domains, focusing on the optimization of business processes, the deployment of pilot digital products, and the redefinition of critical organizational roles.
- **Growth Stage:** This stage is characterized by an institutional focus on scalability and systemic integration. Beyond local pilots, the organization emphasizes the wide-scale adoption of new digital protocols and business models, ensuring that technological advancements are harmonized across all departments.

- **Mature Stage:** In the final stage of transformation, the organization reaches a state of continuous innovation and iterative improvement. A sophisticated framework for digital governance is established, ensuring that the transformation is not only sustained but also remains agile enough to respond to future disruptions.

1-2-Benefits of Digital Transformation

The systemic integration of digital frameworks yields a multitude of high-impact benefits, fundamentally enhancing an organization's competitive posture and operational health:(managed Outsourcing solutions, 2022).

- **Precision and Operational speed:**besides automation, digital integration facilitates increased accuracy by mitigating human error in data-sensitive tasks. This technical reliability, coupled with streamlined workflows, catalyzes higher productivity, allowing the workforce to focus on high-value cognitive activities rather than repetitive administrative burdens.
- **Process Efficiency:** In an increasingly fickle global market, organizational agility becomes a survival mandate. Digital ecosystems enable seamless efficiency in business operations, providing the infrastructure necessary to pivot strategies rapidly and optimize resource allocation in real-time.
- **Catalyzing Opportunities:** Digital maturity acts as a springboard for expansion. By lowering entry barriers to new markets and enabling the rapid deployment of innovative business models, the transformation significantly increases growth opportunities and fosters long-term competitive advantages.
- **Optimization of Stakeholder Engagement:** The transition to digital platforms redefines the interface between the organization and its public. It fosters better customer engagement through personalized, data-informed interactions, leading to higher levels of satisfaction and long-term institutional loyalty.
- **Value Creation:** the synergy of operational savings and market expansion translates into higher revenue and enhanced profitability. By optimizing the cost-to-service ratio, digital transformation ensures the economic sustainability of the organization in the contemporary digital landscape.

2- Electronic management:

2-1- Definition: Within the contemporary information age, Electronic Management transcends mere technological adoption to function as a

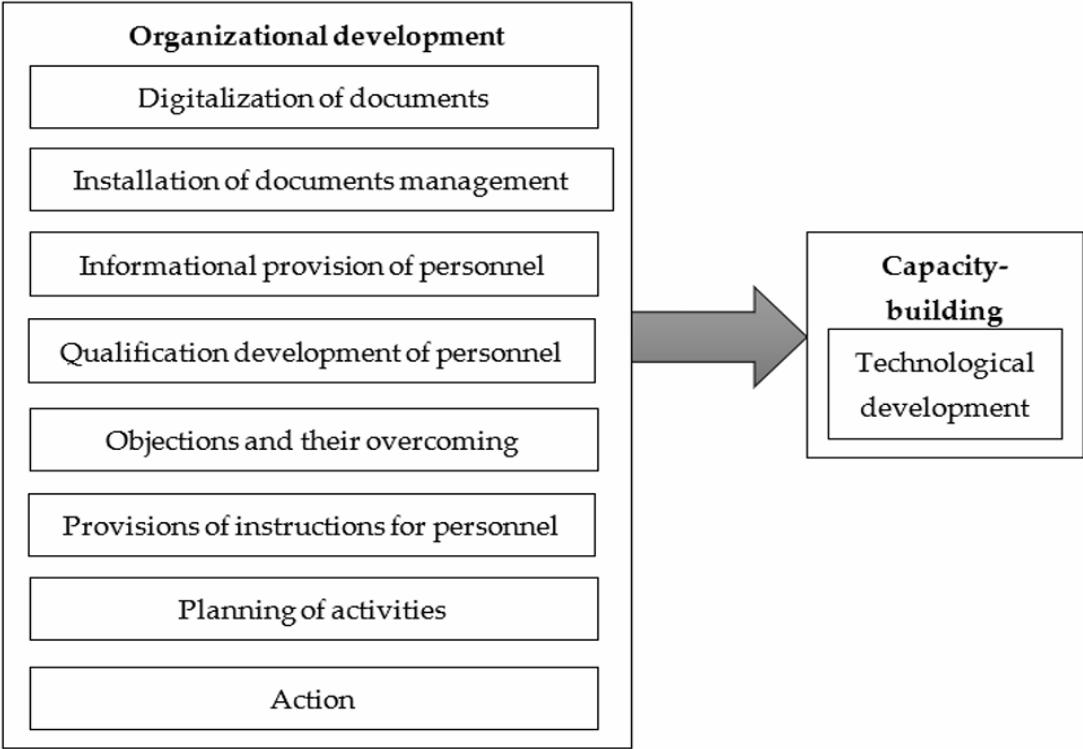
sophisticated administrative strategy. Its primary objective lies in the optimal utilization of information resources within a modernized electronic framework. This strategic approach is meticulously designed to harmonize human and material resources to maximize operational efficiency.

E-Management ensures that organizational goals are met with unprecedented precision. Fundamentally, it represents the systemic capability to leverage information and communication technology (ICT) and cutting-edge techniques to execute complex administrative functions. This transformation enables the seamless migration of traditional activities toward automated computer networks and the internet, establishing a robust, data-driven foundation for modern governance.(ALKHSABAH, 2017, p. 330)

At its core, E-Management is engineered to fortify organizational structures through the systematic development and utilization of enterprise-wide IT projects.

In addition, the objective of E-Management is to cultivate a robust knowledge environment—a digital intelligence layer that empowers the organization to realize its long-term strategic vision with agility and precision.(Liu, Ahmad, Ahmed, & Omar, 2011).Technically, E-Management serves as a sophisticated process that leverages advanced Information and Communication Technologies (ICT) to ensure seamless internal and external organizational functionality. Furthermore, by integrating digital tools into core functions such as marketing and strategic decision-making, E-Management empowers institutions to operate with heightened transparency and data-driven precision. (Neringa & Karolina , 2017).

Figure 01: Thee-management processes in entities.

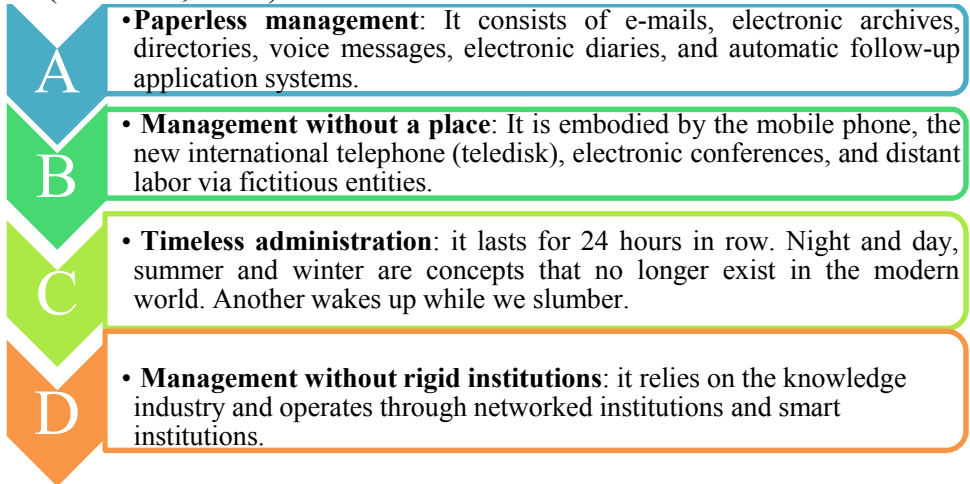


Source:(Neringa & Karolina , 2017).

This foundational phase prioritizes the Digitalization of documents and the systemic Installation of document management systems (DMS), establishing the essential infrastructure for high-velocity information management.

Beyond technical deployment, the success of the system relies on a robust human-centric strategy. This begins with the Informationalprovision of personnel, ensuring that the workforce is seamlessly integrated into the new data environment, followed by Qualification development to enhance the digital competencies of the staff. Crucially, the model accounts for the psychological dimensions of change by addressing Objections and their strategic overcoming, a phase that precedes the dissemination of formal Instructions for personnel.

2-2-Elements of E-Management: Elements of electronic management include: (Basil Ali, 2023).



To put this into action, we will need to: (Basil Ali, 2023)

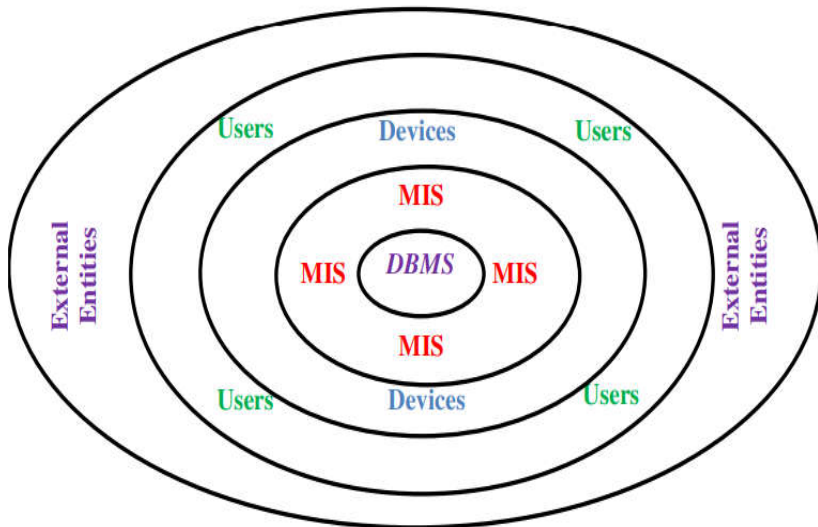
- **Technological Infrastructure:** The physical backbone of the system, comprising the high-performance computing power and hardware architectures necessary to sustain complex digital operations.
- **Software Ecosystems:** A diverse suite of software of all kinds, ranging from specialized management applications to enterprise-wide platforms, designed to automate workflows and secure data integrity.
- **Seamless Connectivity:** The establishment of robust telecommunication networks and protocols that ensure the fluid exchange of information across all institutional levels.
- **Integrated Information Systems:** The structural deployment of information systems that synthesize data streams into actionable intelligence, facilitating centralized control and decentralized execution.
- **Specialized Human Capital:** The recruitment and retention of high-level professional staff—human cadres—who possess the technical expertise required to manage and evolve the digital landscape.
- **Organizational Digital Literacy:** The cultivation of widespread computer awareness and a proactive digital culture among all stakeholders, ensuring that the human element is as agile and prepared as the technological one.

2-3- TheE-Management System?

The Electronic Management System (EMS) is systematically automating business processes. EMS enables companies to achieve substantial service efficiencies and significant time savings.

This transition transcends not merely technical optimization; it represents a fundamental shift in resource management. By delegating repetitive, data-intensive tasks to automated protocols, organizations can mitigate human error and accelerate operational efficiency. Consequently, the primary benefit of an EMS lies in its ability to liberate human capital, allowing professional cadres to refocus their expertise on strategic innovation and high-level decision-making, thereby ensuring long-term institutional sustainability.(Kelli & Joel , 2022).

Figure 02: The layers of E-Management.



Source:(Liu, Ahmad, Ahmed, & Omar, 2011).

Here are the explanations of these layers:

_ Database management system (DBMS): The core or the heart of the whole system is the Enterprise Database, which is a central warehouse where all data is collected, stored, analyzed, and retrieved in an efficient and structured manner.

_ Management information system :(MIS) Management Information System (MIS) is a complete and integrated application system to support the operation of the organization. It is designed on top of E-Management's Principles to achieve E-Management's vision and mission.

_ **Device management:** E-management will utilize all available devices in the industry to increase the level of automation. The devices will interact with the MIS application and Database Management System.

_ **User management:** The fourth layer is all the users who are involved in using the system, either as a process owner or an end-user.

_ **External entity:** The last layer is the external entities of the organization, such as parents, ministry, sponsors, suppliers, customers, banks, or anyone who has interaction with the organization or who needs to access the information from the Enterprise Database.

2-4- The difference between E-Management and E-Government

❖ **E-government:** E-government is one of the electronic management styles and is intended to manage public affairs by electronic means to achieve political, economic, and social goals, and get rid of the routine and the central business, with a high level of transparency. (SAJID , MUATAZ , & HILMI , 2020). Whereas E-management is focused on the internal efficiency and productivity of an organization (whether public or private), utilizing digital tools for better management practices. On the other hand,

❖ E-Government is specifically about how governments use digital technologies to interact with citizens, deliver public services, and manage public administration. While e-management can be a component of e-government, e-government has a broader public-oriented scope that extends beyond internal management to include public service delivery and citizen engagement. (UNSG, 2024).

2-5-The Digital Government Index DGI and digital government maturity:

The DGI assesses digital government maturity along the six dimensions in Figure 3, namely:

1. **Digital by design:** measures how digital government policies are designed to enable the public sector to use digital tools and data coherently when formulating policies or transforming public services.
2. **Data-driven:** measures the government's advancements in developing the governance and enablers needed for data access, sharing, and re-use across the public sector.
3. **Government as a platform:** measures the deployment of common building blocks such as guidelines, tools, data, digital identity, and software to equip teams to advance a coherent transformation of government processes and services across the public sector.

4. **Open by default:** measures openness beyond the release of open data, including efforts to foster the use of technologies and data to communicate and engage with different actors.
5. **User-driven:** measures governments' capacity to place user needs at the core of the design and delivery of public policies and services.
6. **Proactiveness:** measures governments' capacity to anticipate the needs of users and service providers to deliver government services proactively.

Figure 03: The six dimensions of the OECD Digital Government Index



Source:(OECD, 2023).

2-6-Benefits of E-Management

The benefits of E-Management are the great attention given by the developed world using information technology and its various components, caused by the importance and the significant benefits offered by the latter, so countries started to compete in applying electronic management, given the benefits achieved. The following are the most important benefits of electronic administration:(SAJID , MUATAZ , & HILMI , 2020, p. 05)

1) Improve the performance and effectiveness of decision-making by providing information and data to those who want it, facilitating access to them through their presence on the internal network, and accessing them with minimal effort using available search mechanisms.

2) Flexibility in the employee's work so that the employee can easily access the internal network from anywhere may exist.

3) **Ease of meetings remotely**: to hold and attend a meeting from outside the office.

4) There will be no need for a large number of file cabinets, thus providing an area as well as providing a dedicated employee to take care of these files' employee needs.

5) Easy and fast access to instructions and transactions for administrative staff, customers, and reviewers.

6) Ease the ending reviewers' transactions through a single point to do the job on behalf of the other departments.

7) Ease of storage and archiving of data and information, and protection from natural disasters and other factors by keeping backups in places outside the enterprise boundaries.

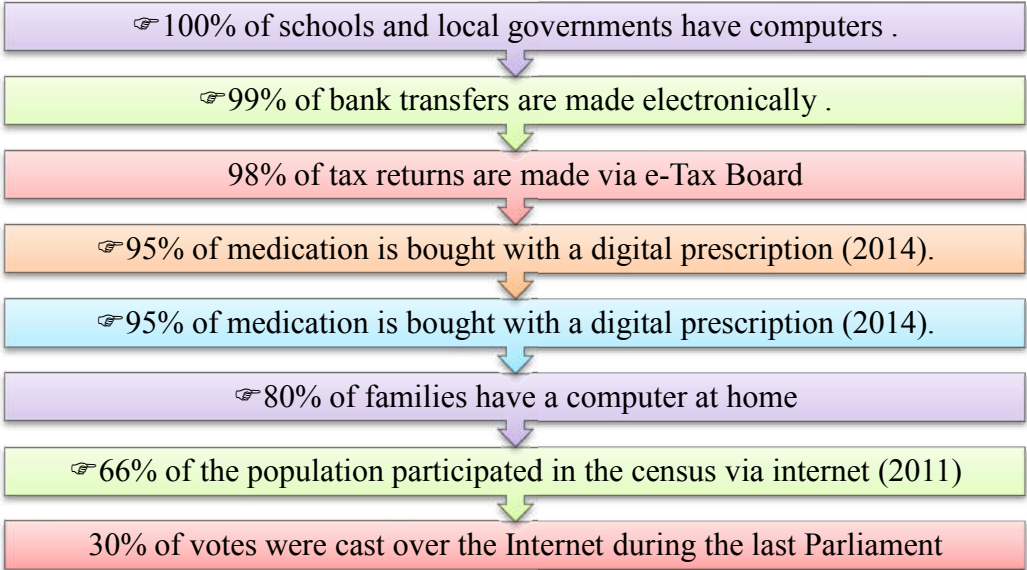
3-Road System as A case study in Estonia:

3-1-Overview of the road system: Estonia is one of the most advanced e-societies in the world. An incredible success story that links anongoing-thinking government, a proactive ICT sector, and a tech-savvy population. Estonia has the world's leading IT infrastructure and e-services.(Sandra & Raul, 2016).

The Baltic states, due to the 99% of public services they continuously maintain, consistently rank among the highest in public trust in governments in the EU. All public services except for marriage and divorce can be accessed online by citizens through this digital transformation.Overall, the Estonian government claims that the digitization of public services saves the country more than 1,400 years of working time and 2% of its GDP annually.(Techforgood, 2022).

The Estonian e-government infrastructure and its success rest on two main pillars, both introduced in 2001, that essentially create a digital state and digital citizens: the data infrastructure X-Road and a compulsory national digital ID.(Rainer & Ines , 2019).

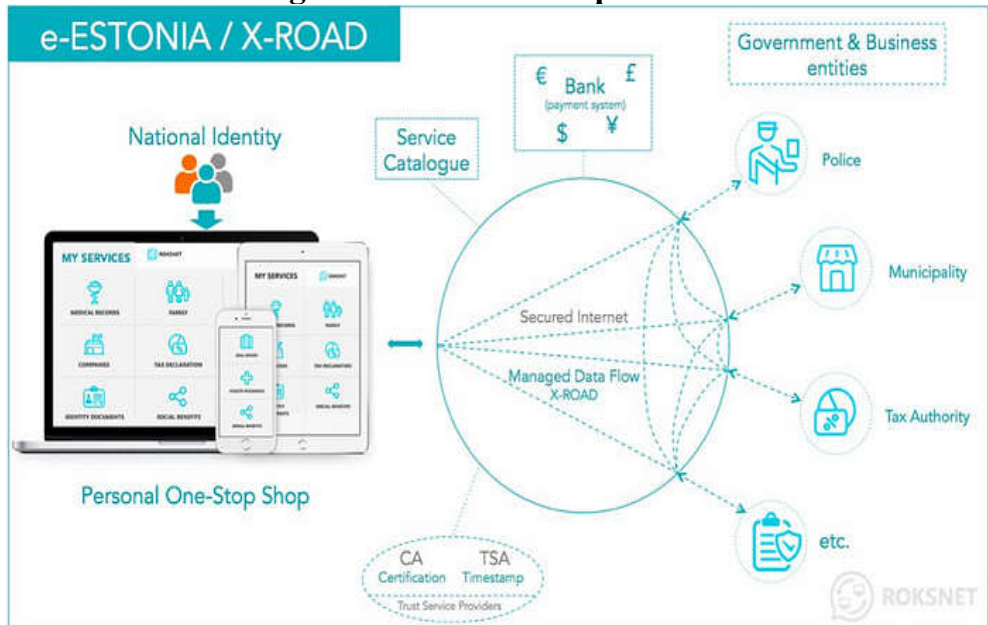
Figure 04: Estonian e-Society indicators:



Source: developed by the authors according to (Sandra & Raul, 2016).

3-2 -X-Road System function:

❖ **What is x-road:** x-road® is an open-source software and ecosystem solution that provides unified and secure data exchange between organizations. Also, X-Road provides a robust and secure solution for exchanging data and establishing a collaborative ecosystem. It streamlines data exchange procedures, enhances security, and facilitates interoperability, which enables organizations to derive greater value from their data assets. (x-road, 2024).

Figure04: E-Esstonia platform.

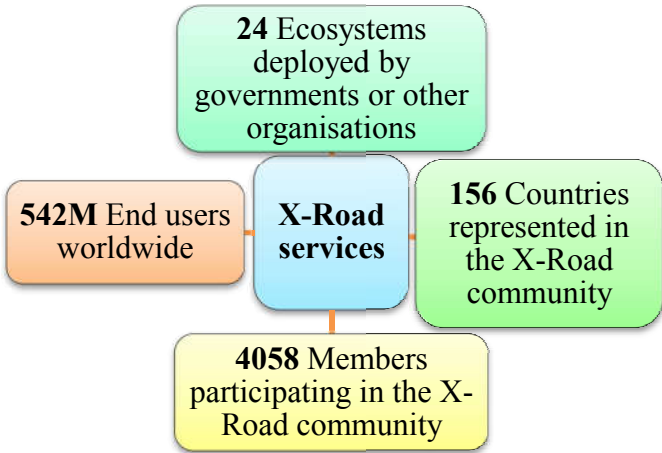
Source:(Mustafa, 2020).

X-Road is a data exchange layer between service consumers and business services provided by various information systems owned by different organizations.

❖ **X-Road services:**X-Road implements a set of common features to support and facilitate data exchange. It provides the following features out of the box:(x-road, 2024).

- ☞ address management.
- ☞ message routing
- ☞ access rights management
- ☞ Organizational-level authentication
- ☞ machine-level authentication
- ☞ transportation layer encryption
- ☞ time-stamping
- ☞ digital signature of messages
- ☞ logging
- ☞ error handling.

Figure05: X-Roadfeatures in Estonia.

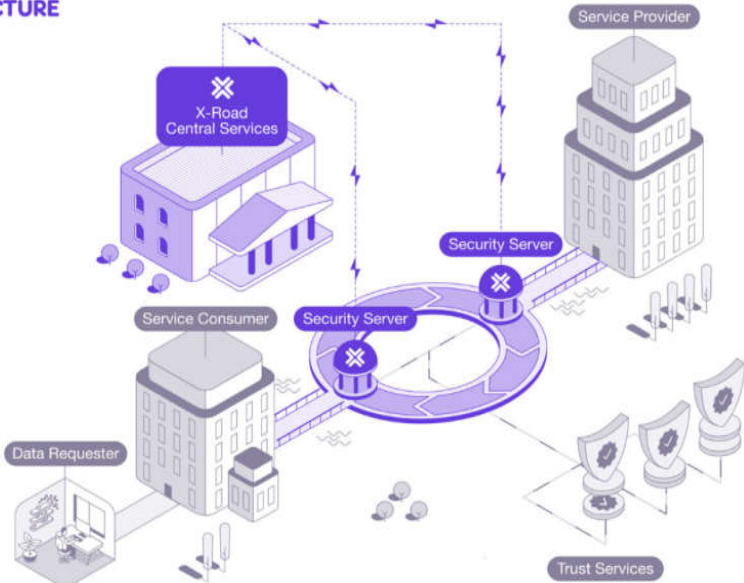


Source: developed by the authors according to (x-road, 2024).

Figure 06: X-Road networks.

Technically, the X-Road ecosystem consists of Central Services, Security Servers, Information Systems, TSA(s), and CA(s)

X-ROAD ARCHITECTURE

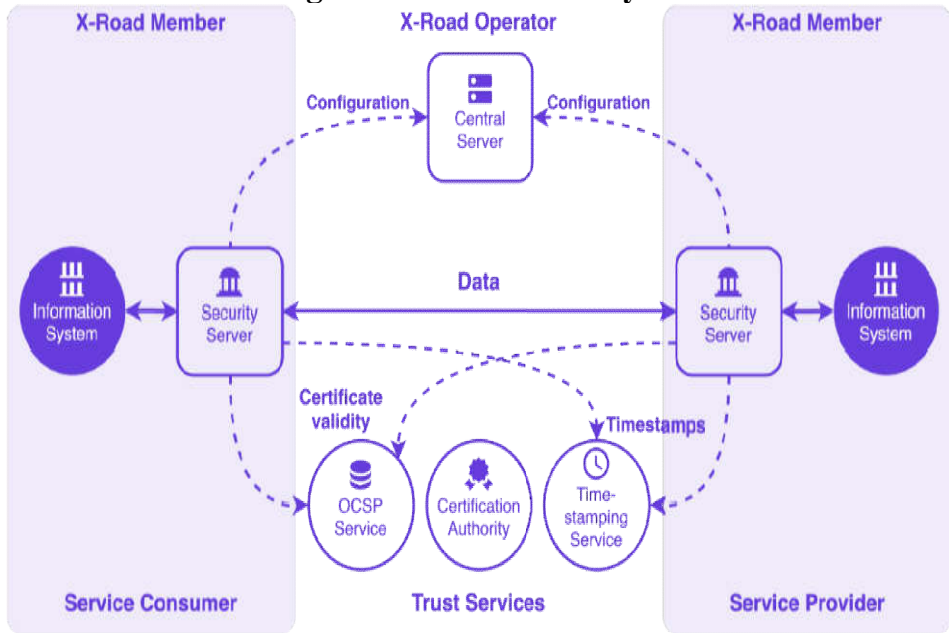


Resource: (x-road, 2024).

X-Road is a distributed data exchange system between Information Systems (IS) within the trusted network. It relies on Key Management Mechanisms PKI to ensure trust between network members. Members are

organizations, with their ISs used for operations by internal users (e.g., employees) and external users (e.g., customers). For cross-organizational data exchange, a component called Security Server (SS) serves as an intermediary. (Mariia & Jan, 2024).

Figure 07: X-Road ecosystem.



Resource: (x-road, 2024).

A data transaction between a service consumer and a service provider consists of the following steps: (x-road, 2024)

- The service consumer information system initiates a data transaction by sending a request to the service consumer Security Server.
- The service consumer Security Server signs and logs the request and routes it to the service provider Security Server. The connection between the Security Servers is secured using mutual TLS (mTLS).
- The service provider Security Server verifies and logs the request and forwards it to the service provider information system.
- The information system processes the request and returns a response to the service provider's Security Server.
- The service provider's Security Server signs and logs the response and forwards it to the service consumer's Security Server. The connection between the Security Servers is secured using mTLS.
- The service consumer Security Server verifies and logs the response and forwards it to the service consumer information system.
- The service consumer information response.

- X-Road connects different information systems that may include a variety of services. It has developed into a tool that can also write to multiple information systems, transmit large data sets, and perform searches across several information systems simultaneously. X-Road was designed with growth in mind, so it can be scaled up as new e-services and platforms come online.

3-3-X-Road benefits for organizations:

- **Centralized governance:** X-Road provides a centrally governed distributed platform for managing data exchange within an ecosystem.

- **Scalability:** X-Road is designed to handle large-scale data exchange ecosystems.

- **Trust and transparency:** X-Road promotes trust among ecosystem participants by ensuring secure and auditable data exchange.

- **Collaboration opportunities:** X-Road fosters collaboration among participating organizations by enabling seamless data sharing.

- **Cost savings:** By deploying X-Road, organizations can avoid the development costs of building custom data exchange solutions. It offers a reusable solution, reducing the overall investment required for setting up a data exchange ecosystem.

6-4 - X-Road benefits for companies

- **Market potential:** X-Road is a widely recognized and trusted data exchange solution that organizations and governments use worldwide. A company can tap into a market that values secure and efficient data exchange solutions by offering projects that utilize X-Road. This provides an opportunity for revenue generation and business growth.

- **Competitive advantage:** Leveraging X-Road as the data exchange solution gives a company a competitive edge. It demonstrates the company's commitment to data security, interoperability, and adherence to international best practices. This can differentiate the company's projects from competitors and attract potential clients who prioritize secure and unified data exchange.

- **Reusability:** X-Road is an open-source software solution, which means it can be customized and adapted to meet specific project requirements. A company can leverage the flexibility of X-Road to develop projects tailored to clients' unique needs.

- **Cost savings and efficiency:** A company can save development costs and time by utilizing X-Road. Instead of building a data exchange solution from scratch, the company can benefit from the existing infrastructure and components of X-Road. This leads to faster project

implementation, reduced development effort, and cost savings for the company and its clients.

➤ **Support and community resources:** X-Road has an active community of users, developers, and contributors. Companies selling projects utilizing X-Road can benefit from the support and resources available within the community. They can access documentation, forums, and peer-to-peer expert advice to help in project development, troubleshooting, and ongoing maintenance.

➤ **X-Road Technology Partner Program:** Enterprises providing consultation services, e.g., deploying independent instances, developing extensions and compatible services, integrating information systems, etc., can apply to become part of the X-Road Technology Partner Program.

4-Conclusion:

X-Road's implementation in Estonia has saved substantial time and cost, improved efficiency in public service delivery, enhanced transparency, and contributed to Estonia's reputation as a digital pioneer. The platform's success is measurable not only in economic terms but also in the level of trust and participation among citizens and businesses. For example, the Government of Estonia notes that digitalization of public services saves the country more than 2 percent of its GDP annually, and digital service delivery saves Estonians the equivalent of more than 1,400 years of working time every year due to more efficient service delivery.(UNDP, 2024).

In addition, selling or buying a car in Estonia can be done remotely in less than 15 minutes, paying off an online tax takes an average person no more than five minutes, and participating in elections by Internet voting takes 90 seconds on average.(Vassil, 2016).

5. Bibliography List:

1. Managed Outsourcing solutions. (2022, 08 29). What are the 3 Main Components of Digital Transformation and How Should I Implement It? Retrieved 09 22, 2023, from <https://www.managedoutsourcing.com/blog/3-main-components-of-digital-transformation-i-should-implement/>
2. ALKHSABAH, M. A. (2017, 01). Reality of Use of Electronic Management and Its Impact on the Job. *International Journal of Academic Research in Accounting, Finance and Management Sciences*, pp. 329–341. Retrieved 08 22, 2024, from https://hrmars.com/papers_submitted/2724/Article_34_Reality_of_Use_of_Electronic_Management.pdf
3. Basīl Ali, H. (2023, 03 01). The role of e-Management in the enterprise resource planning case study on the Industry Sector in Iraq. *Journal for Educators, Teachers and Trainers*, 14(02), pp. 25-35. doi: 10.47750/jett.2023.14.02.003
4. COURSERA. (2023, 06 16). What Is Digital Transformation? Definition, Strategy, and Examples. Retrieved 08 01, 2023, from <https://www.coursera.org/articles/digital-transformation>
5. David, E., & Sechi, K. (2022, 01). The 2021 Digital Services Convening. Retrieved 08 21, 2024, from https://ash.harvard.edu/wp-content/uploads/2024/02/digital_services_convening_2021.pdf
6. Kelli, H., & Joel, T. (2022, 05 24). What Is an Electronic Management System? Retrieved 08 20, 2024, from <https://www.unleash.so/https://www.unleash.so/a/answers/knowledge-management/what-is-an-electronic-management-system>
7. Liu, Y., Ahmad, B. O., Ahmed, N. A., & Omar, R. M. (2011, 08 18). E-management development and deployment strategy. *African Journal of Business Management*, 6657-6667, pp. 6657-6667. doi:10.5897/AJBM10.1594
8. Mariia, B., & Jan, K. (2024). The Power of Many: Securing Organisational Identity Through Distributed Key Management. *springer*, 08(21), . 475–491. Retrieved from https://link.springer.com/chapter/10.1007/978-3-031-61057-8_28
9. Mustafa, A. (2020, 11 02). Digital Government: Design, Development, and Evolution of Estonia's X-Road. Retrieved 08 21, 2024, from <https://arifmustafa.medium.com/https://arifmustafa.medium.com/digital-government-design-and-development-and-evolution-of-estonias-x-road-78d4dc01e01f>
10. Neringa, V.-V., & Karolina, P. (2017). E-Management as a Game Changer in Local Economies, 10. Retrieved 08 22, 2024, from <https://www.mdpi.com/2227-7099/10/8/180>
11. OECD. (2023). 2023 OECD Digital Government Index Results and Key Findings. Paris: OECD Publishing.
12. Rainer, K., & Ines, M. (2019, 09). Estonia's Digital Transformation: Mission Mystique and the Hiding Hand. *great policy success*, pp. 143–160. Retrieved from <https://doi.org/10.1093/oso/9780198843719.003.0008>

13. SAJID , R., MUATAZ , K., & HILMI , H. (2020, 01). INTERNATIONAL JOURNAL OF RESEARCH IN REGIONAL STUDIES, LAW, SOCIAL, pp. 2455-6602. Retrieved 08 22, 2024, from <https://publications.anveshanaindia.com/wp-content/uploads/2020/01/A-STUDY-ON-ELECTRONIC-MANAGEMENT-AND-ITS-ROLE-IN-MANAGERIAL-FUNCTIONS.pdf>
14. Sandra, R., & Raul, R. (2016, 12 30). e-Estonia: e-Governance in Practice. (N. G. Program, Ed.) Retrieved 08 21, 2024, from <https://ega.ee>: <https://ega.ee/wp-content/uploads/2016/06/e-Estonia-e-Governance-in-Practice.pdf>
15. Techforgood. (2022, 01 05). Exploring e-Estonia. Retrieved 08 20, 2024, from <https://www.techforgood.net/articles/exploring-e-estonia>: <https://www.techforgood.net/articles/exploring-e-estonia>
16. UNDP. (2024, 03). Return-on-Investment in National Digital Transformation: Retrieved 08 21, 2024, from <https://www.undp.org>: <https://www.undp.org/sites/g/files/zskgke326/files/2024-04/undp-dfs-roi-in-national-digital-transformation-exploring-the-development-impact-of-digital.pdf>
17. UNSG. (2024). PROMOTING DIGITAL PUBLIC GOODS: THE UNITED NATIONS SECRETARY-GENERAL. Retrieved 08 21, 2024, from <https://www.un.org/>: https://www.un.org/techenvoy/sites/www.un.org.techenvoy/files/general/Digital_Public_Goods_Summary_PDF.pdf
18. Vassil, K. (2016). Estonian e-Government Ecosystem Foundation, Applications, Outcomes. World Development Report. Retrieved 08 21, 2024, from <https://thedocs.worldbank.org/en/doc/165711456838073531-0050022016/original/WDR16BPEstonianeGovecosystemVassil.pdf>
19. x-road. (2024). X-Road® Architecture X-Road ecosystem consists of Central Services, Security Servers, Information Systems, TSA(s), and CA(s). Retrieved 08 21, 2024, from <https://x-road>: <https://x-road.global/>